

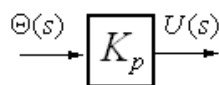
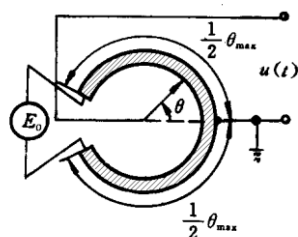
§ 2.3.2 常用控制元件的传递函数

(1) 电位器

$$\frac{u}{\frac{E_0}{2}} = \frac{\theta}{\frac{\theta_{\max}}{2}}$$

$$u = \frac{E_0}{\theta_{\max}} \cdot \theta = K_p \theta$$

$$G(s) = \frac{U(s)}{\Theta(s)} = K_p$$

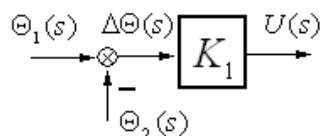
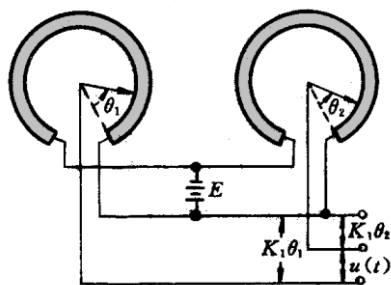


(2) 电桥式误差角检测器

$$\begin{cases} u_1 = K_1 \theta_1 \\ u_2 = K_1 \theta_2 \end{cases}$$

$$u = u_1 - u_2 = K_1(\theta_1 - \theta_2) = K_1 \cdot \Delta\theta$$

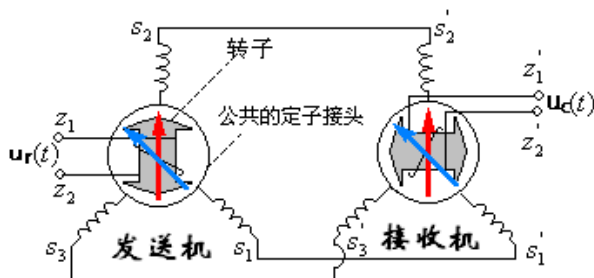
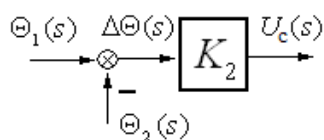
$$G(s) = \frac{U(s)}{\Delta\Theta(s)} = K_1$$



(3) 自整角机

$$u_c = K_2 \sin(\theta_1 - \theta_2) \approx K_2 \cdot \Delta\theta$$

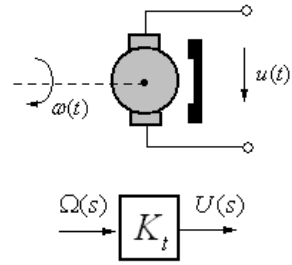
$$G(s) = \frac{U_c(s)}{\Delta\Theta(s)} = K_2$$



(4) 直流测速发电机

$$u = K_t \cdot \omega$$

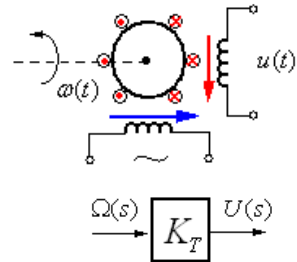
$$G(s) = \frac{U(s)}{\Omega(s)} = K_t$$



(5) 交流测速发电机

$$u = K_T \cdot \omega$$

$$G(s) = \frac{U(s)}{\Omega(s)} = K_T$$

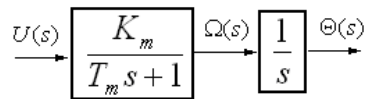


(6) 电枢控制式直流电动机

$$T_m \dot{\omega} + \omega = T_m \ddot{\theta} + \dot{\theta} = K_m u$$

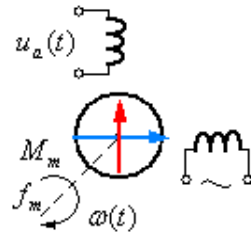
$$T_m s \Omega(s) + \Omega(s) = T_m s^2 \Theta(s) + s \Theta(s) = K_m U(s)$$

$$\begin{cases} G_\Omega(s) = \frac{\Omega(s)}{U(s)} = \frac{K_m}{T_m s + 1} \\ G_\theta(s) = \frac{\Theta(s)}{U(s)} = \frac{K_m}{s(T_m s + 1)} \end{cases}$$



(7) 两相异步电动机

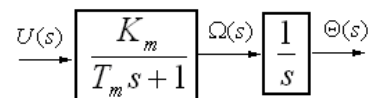
$$\begin{cases} \text{堵转力矩} & M_s(t) = c_m \cdot u_a(t) \\ \text{机械特性} & M_m(t) = M_s(t) - c_\omega \cdot \omega(t) \\ \text{牛顿定律} & J_m \dot{\omega}(t) + f_m \omega(t) = M_m(t) \end{cases}$$



●方法1 消去中间变量 ($M_s(t)$, $M_m(t)$) 得:

$$T_m \dot{\omega}(t) + \omega(t) = K_m \cdot u_a(t)$$

$$\begin{cases} T_m = J_m / (f_m + c_\omega) \\ K_m = c_m / (f_m + c_\omega) \end{cases} \quad G(s) = \frac{\Omega(s)}{U_a(s)} = \frac{K_m}{T_m s + 1}$$



- 方法 2 对子方程分别进行拉氏变换

$$\left\{ \begin{array}{l} \text{堵转力矩} \quad M_s(t) = c_m \cdot u_a(t) \end{array} \right. \quad (1)$$

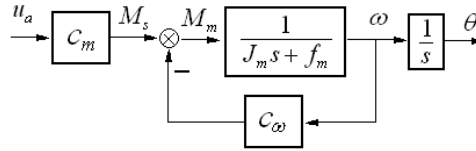
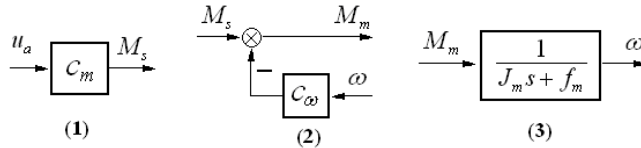
$$\left\{ \begin{array}{l} \text{机械特性} \quad M_m(t) = M_s(t) - c_\omega \cdot \omega(t) \end{array} \right. \quad (2)$$

$$\left\{ \begin{array}{l} \text{牛顿定律} \quad J_m \dot{\omega}(t) + f_m \omega(t) = M_m(t) \end{array} \right. \quad (3)$$

$$\left\{ \begin{array}{l} M_s(s) = c_m \cdot U_a(s) \end{array} \right. \quad \frac{M_s(s)}{U_a(s)} = c_m \quad (1)$$

$$\left\{ \begin{array}{l} M_M(s) = M_s(s) - c_\omega \cdot \Omega(s) \end{array} \right. \quad (2)$$

$$\left\{ \begin{array}{l} J_m s \cdot \Omega(s) + f_m \Omega(s) = M_M(s) \end{array} \right. \quad \frac{\Omega(s)}{M_M(s)} = \frac{1}{J_m s + f_m} \quad (3)$$

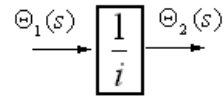


两相异步电动机结构图

(8) 齿轮系 (传动比 $i = \frac{z_2}{z_1} = \frac{\omega_1}{\omega_2}$)

$$\omega_2 = \frac{1}{i} \cdot \omega_1$$

$$G(s) = \frac{\Theta_2(s)}{\Theta_1(s)} = \frac{\Omega_2(s)}{\Omega_1(s)} = \frac{1}{i}$$



$$\left\{ \begin{array}{l} J_{\text{总}} = J_1 + \left(\frac{z_1}{z_2}\right)^2 \cdot J_2 + \left(\frac{z_1}{z_2} \cdot \frac{z_3}{z_4}\right)^2 \cdot J_3 + \dots \\ f_{\text{总}} = f_1 + \left(\frac{z_1}{z_2}\right)^2 \cdot f_2 + \left(\frac{z_1}{z_2} \cdot \frac{z_3}{z_4}\right)^2 \cdot f_3 + \dots \end{array} \right.$$

